SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA04 COURSE CODE: Operating Systems

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 5 **Total Marks:25**

Annexure A

Concept Mapping

Code of Conduct:

I Vishwa.U(192511192)___(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Concept Mapping

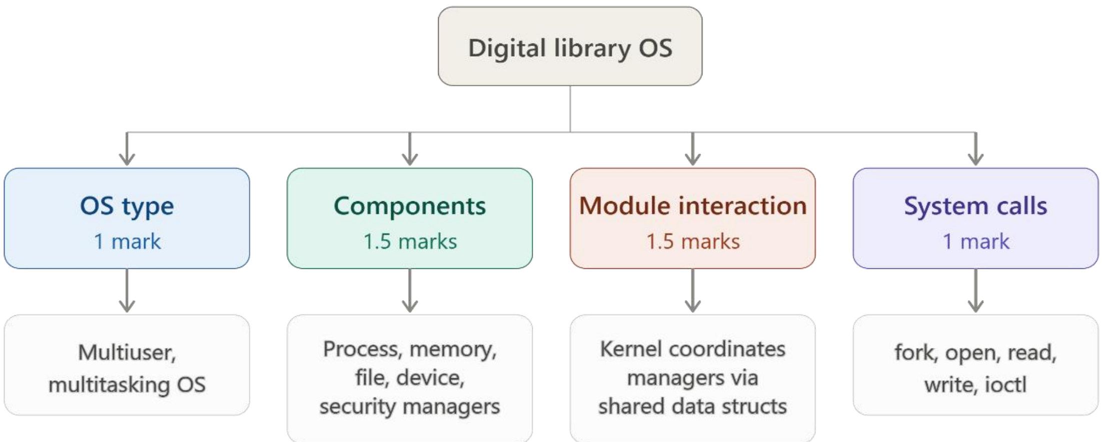
Question 1 (5 Marks)
Construct a multi-level concept map integrating:
Operating System Types → OS Structures → System Calls → OS Functions.
Your map must show cross-links (e.g., how OS structure influences system call implementation).

Question 2 (5 Marks)
Design a comprehensive concept map representing:
Process Management, Memory Management, File System, and I/O System.
Include interdependencies and indicate where system calls interact with each component.

Question 3 (5 Marks)
Create a comparative concept map for:
Batch OS, Time-Sharing OS, Distributed OS, and Real-Time OS.
Highlight differences, similarities, advantages, and limitations using linking phrases.

Question 4 (5 Marks)
Develop a dynamic concept map showing the execution flow:
User Program → API → System Call Interface → Kernel → Hardware → Response back to User.
Include feedback loops and control flow paths.

Question 5 (5 Marks)
Construct a hierarchical + functional concept map for system calls:
Process, File, Device, and Information system calls.
Also show how they relate to OS services like scheduling, memory allocation, and device handling.



Page 1: OS type & components

OS type & justification

- Multiuser & multitasking
- Time-shared / network OS
- Justification:
  many concurrent users,
  need reliability, security

Major components/services

- Process mgmt
- Memory mgmt
- File mgmt
- Device mgmt
- Security / communication mgmt

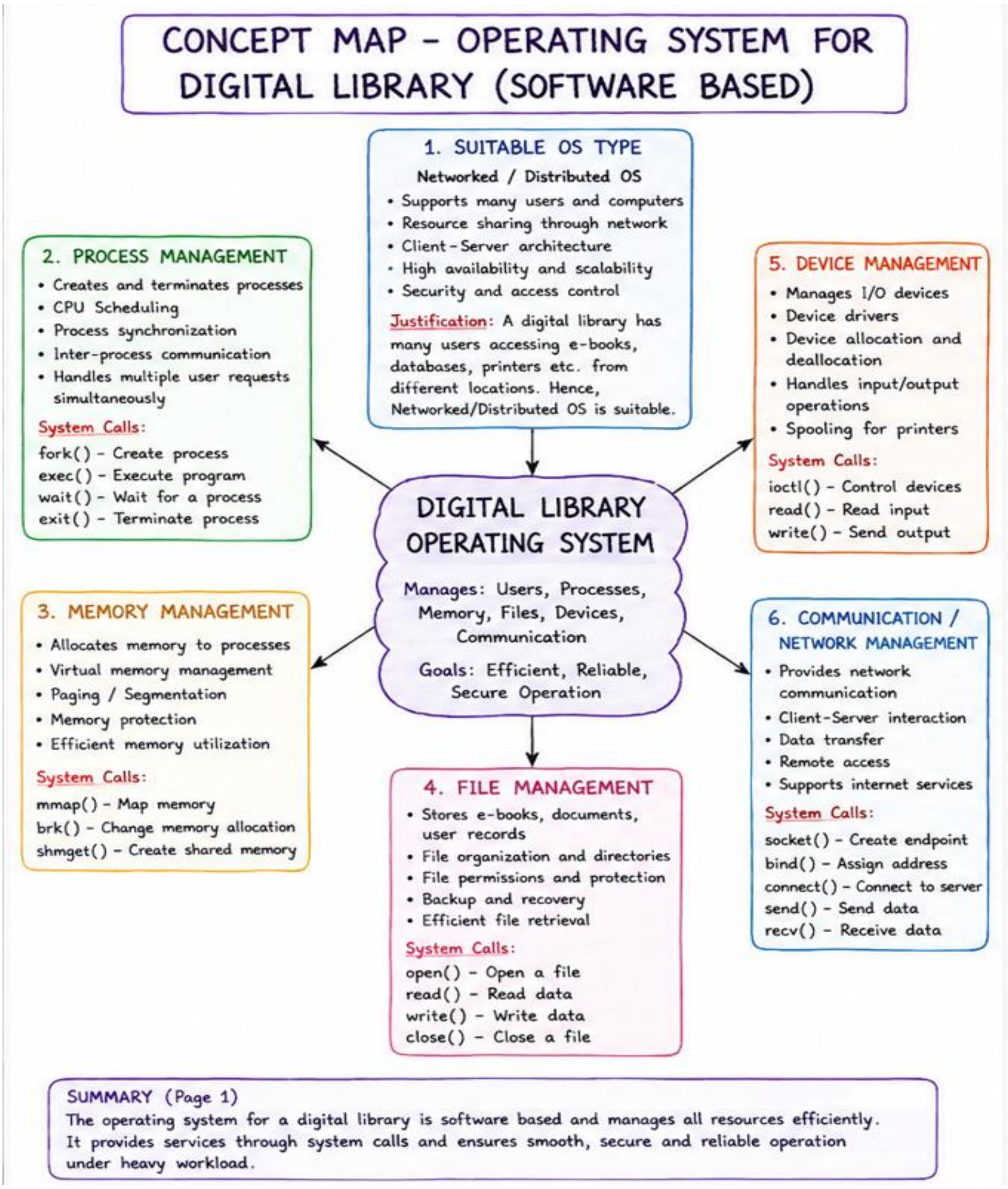
Page 2: interaction & system calls

Module interaction

- Kernel coordinates managers
- Shared kernel data structures
- Example:
  process manager requests
  pages from memory manager

Relevant system calls

- fork() — new user session
- open()/read() — fetch book file
- write() — save borrow record
- ioctl() — configure scanner/printer


CONCEPT MAP - OPERATING SYSTEM FOR DIGITAL LIBRARY (SOFTWARE BASED)
1. SUITABLE OS TYPE
Networked / Distributed OS
• Supports many users and computers
• Resource sharing through network
• Client-Server architecture
• High availability and scalability
• Security and access control
Justification: A digital library has many users accessing e-books, databases, printers etc. from different locations. Hence, Networked/Distributed OS is suitable.
2. PROCESS MANAGEMENT
• Creates and terminates processes
• CPU Scheduling
• Process synchronization
• Inter-process communication
• Handles multiple user requests simultaneously
System Calls:
fork() - Create process
exec() - Execute program
wait() - Wait for a process
exit() - Terminate process
5. DEVICE MANAGEMENT
• Manages I/O devices
• Device drivers
• Device allocation and deallocation
• Handles input/output operations
• Spooling for printers
System Calls:
ioctl() - Control devices
read() - Read input
write() - Send output
DIGITAL LIBRARY OPERATING SYSTEM
Manages: Users, Processes, Memory, Files, Devices, Communication
Goals: Efficient, Reliable, Secure Operation
3. MEMORY MANAGEMENT
• Allocates memory to processes
• Virtual memory management
• Paging / Segmentation
• Memory protection
• Efficient memory utilization
System Calls:
mmap() - Map memory
brk() - Change memory allocation
shmget() - Create shared memory
6. COMMUNICATION / NETWORK MANAGEMENT
• Provides network communication
• Client-Server interaction
• Data transfer
• Remote access
• Supports internet services
System Calls:
socket() - Create endpoint
bind() - Assign address
connect() - Connect to server
send() - Send data
recv() - Receive data
4. FILE MANAGEMENT
• Stores e-books, documents, user records
• File organization and directories
• File permissions and protection
• Backup and recovery
• Efficient file retrieval
System Calls:
open() - Open a file
read() - Read data
write() - Write data
close() - Close a file
SUMMARY (Page 1)
The operating system for a digital library is software based and manages all resources efficiently. It provides services through system calls and ensures smooth, secure and reliable operation under heavy workload.

CONCEPT MAP (CONTINUED) - PAGE 2

7. SECURITY AND PROTECTION

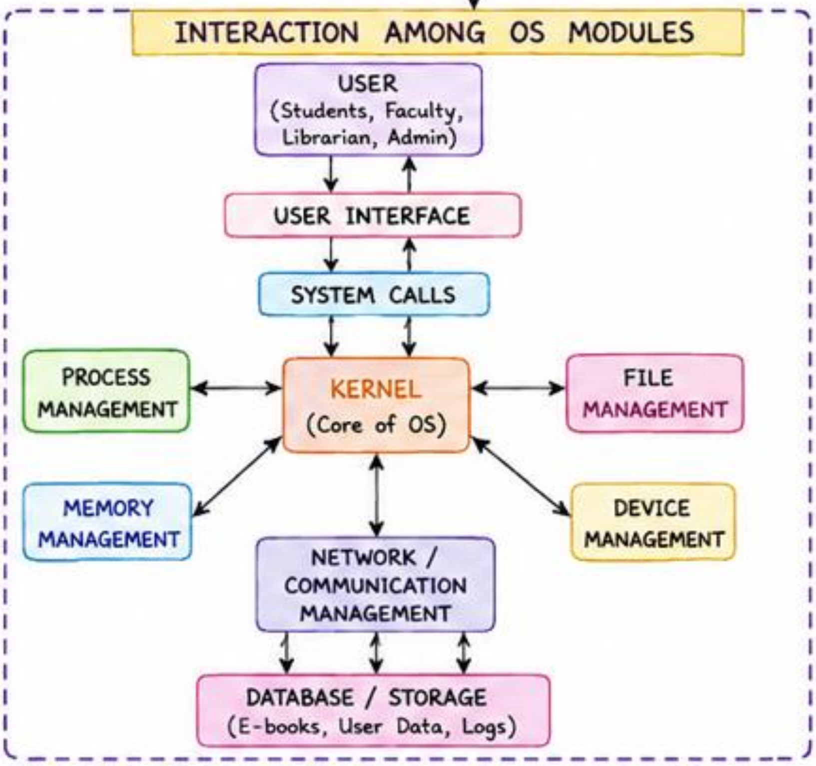
- User authentication
- Access control and authorization
- File permissions (Read/Write/Execute)
- Data encryption
- Protects user privacy
- Prevents unauthorized access

System Calls:

getuid() - Get user ID
setuid() - Set user identity
chmod() - Change file permissions

8. USER INTERFACE (UI)

- Easy interaction for users
- Search books
- Read / Download e-books
- Issue / Return digital resources
- Notifications and alerts
- Multilingual support
- User friendly design



IMPORTANT SYSTEM CALLS

Process Management

fork(), exec(), wait(), exit()

Memory Management

mmap(), brk(), shmget()

File Management

open(), read(), write(), close()

Device Management

ioctl(), read(), write()

Communication

socket(), bind(), connect(), send(), recv()

Security

getuid(), setuid(), chmod()

SYSTEM CALL FLOW

User Request → System Call → Kernel (OS) → OS Modules (Process, Memory, File, Device, Network) → Resource Access → Result / Response to User

CONCLUSION

A software based operating system with proper modules and system calls can efficiently manage users, processes, memory, files, devices and communication in a digital library, ensuring reliable, secure and efficient service under heavy workload.

RUBRICS FOR CALCULATION

Concept Mapping					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand